

# Grounding the Robo-Advisor: 110 – Age vs. the Research

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| Client                                                         | Heuristic (110 – age) | MVO   | Research-informed |
|----------------------------------------------------------------|-----------------------|-------|-------------------|
| A: age 35, Moderate, 30y, \$100,000, \$2,000/mo                | 75.0%                 | 64.2% | 90.4%             |
| B: age 68, Moderate, 20y, \$1,500,000, \$0/mo or \$48k pension | 42.0%                 | 64.2% | 60.8%             |

## 1. What changed?

Before grounding, the only lifecycle rule was the popular 110 – age heuristic. After grounding in Duarte, Fonseca, Goodman & Parker (2021) and Choi, Liu & Liu (2025), the research mode moved Client A up 15.4 percentage points (75.0% to 90.4%) and Client B up 18.8 percentage points (42.0% to 60.8%).

Age alone cannot explain A's jump. The Duarte glide at 35 is 84.0%, only 9 points above the heuristic. The rest comes from the ratio of human capital to financial wealth: A's future wages are worth  $H = \$3,212,024$  against  $W = \$100,000$ , so  $H/W = 32.12$ . The Merton share of total wealth is  $\alpha^* = 41.0\%$ , and scaling it by  $(1 + H/W)$  drives the financial-wealth share  $\hat{\alpha}$  to its 100.0% cap. The 60/40 blend of Duarte and  $\hat{\alpha}$  gives  $0.6 \times 84.0\% + 0.4 \times 100.0\% = 90.4\%$ .

For B, the pension-like income is worth  $H = \$764,972$  against  $W = \$1,500,000$ , so  $H/W = 0.51$ .  $\alpha^* = 41.0\%$  scales to  $\hat{\alpha} = 61.9\%$ , the Duarte retirement level is 60.0%, and the blend is  $0.6 \times 60.0\% + 0.4 \times 61.9\% = 60.8\%$ . MVO did not move because it targets an 11% volatility for a Moderate client and never sees age or income, so it returns 64.2% for both.

## 2. Was it an improvement?

For Client B, yes. A 68-year-old with a 20-year horizon and a pension is not a 42% equity investor, and a typical target-date fund that drifts to 30–40% is worse still. Duarte et al. find the optimal equity share of financial wealth settles at a stable ~60% in retirement and stays there, and they estimate that age-only glides cutting to 30–40% cost 2–3% of consumption. Choi, Liu & Liu add the mechanism: Social Security and pension income is bond-like, so the safe asset is already on the household balance sheet and the financial account does not have to supply it. Mode C's 60.8% sits inside the 55–65% band that the literature implies for a funded retiree.

For Client A, the direction is defensible but 90.4% is not obviously right. The model treats A's wages as a \$3.2 million bond, which is true only if the job is secure and the money is never needed early. Sequence-of-returns risk near retirement, job loss correlated with a market drop, and near-term spending such as a home purchase are outside the model. Those are judgment calls for a human advisor, not outputs of a spreadsheet.

## 3. What did the AI get wrong?

Four things, each caught by checking rather than by trusting the first draft. Retirement age was hardcoded at 65 until we moved it to 67, which shifts both the Duarte path and how long human capital accrues. A retiree who entered \$0 income had  $H$  wiped to zero, which collapses  $\hat{\alpha}$  back to  $\alpha^*$  and drops the recommendation; the pension and Social Security stand-in fixed that. A naive TDF glide would have put B at 30–40%, the number Duarte identifies as too low. And raw Practical Finance saturates A at  $\hat{\alpha} = 100.0\%$ , which no adviser would sign; the blend with the Duarte glide is what pulled A back to 90.4%.

## 4. The meta-lesson

An ungrounded assistant reproduces 110 – age because that is what the training data says most often. Grounding did not just change a coefficient. It changed the state variables the model conditions on, from age alone to age plus  $H$  and  $W$ , and it changed the retirement level from a declining glide to a flat ~60%. That makes Mode C a better first draft than the heuristic. It does not make it a safer last draft. The 55–65% band for Client B was a unit test, chosen because the papers imply it, not a statement of faith in the model. When the first pass failed that test, the fix was in the assumptions, not in the papers.

**References.** Duarte, V., Fonseca, J., Goodman, A. S., & Parker, J. A. (2021). *Simple Allocation Rules and Optimal Portfolio Choice Over the Lifecycle*. NBER Working Paper 29559. Choi, J. J. (2022). Popular Personal Financial Advice versus the Professors. *Journal of Economic Perspectives*, 36(4). Choi, J. J., Liu, S., & Liu, Y. (2025). *Practical Finance*. NBER Working Paper 34166.